



Burns & Chronic Wounds

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This lecture will discuss burn presentations and their initial management; Subsequent care of critically-ill burn patients is not discussed.

Will also briefly discuss chronic wound management.

Trigger Warning

- Please know that included in this lecture are some graphic images of burns as well as discussion of abuse.
- These images and information, although sensitive and potentially triggering, are important to the overall discussion and therefore were consciously included.
- Feel free to step out, take breaks, or otherwise excuse yourself at any time as you need it.

Objectives

- ◆ Define Burn Classifications
- ◆ Estimate extent of a burn using the “Rule-of-Nines” (necessary for resuscitation)
- ◆ Describe treatment of burns using the "6 C's"
- ◆ Understand the tenets of chronic wound care
- ◆ Describe criteria for staging pressure ulcers
- ◆ Understand staging for diabetic foot ulcers

Epidemiology of Burns

- ◆ Each year, roughly 450,000 people suffer from burn injuries that require treatment
 - ◆ A burn injury every 70 seconds
- ◆ Children ages < 5 and older adults are at the greatest risk for severe injury
- ◆ *An overwhelming majority of scald burns occur in the home*



Reference: [American Burn Association](#) (article)

Reference: [Thermal Burns](#) (article)

BURNS

- ◆ Peds most common: scald injury (hot liquids, cooking oils, steam)
- ◆ Adults most common: flame
- ◆ Depth of burn injury is related to:
 - contact temperature
 - duration of contact of the external heat source
 - thickness of skin (region of body)



Types of Skin Injuries

- ◆ Thermal burns - MOST COMMON TYPE OF BURN

- ◆ Thermal conductivity of skin is low, so most thermal burns only involve the epidermis and part of the dermis

- ◆ Flames, hot liquids (water, grease), steam

- ◆ Cold exposure (frostbite)

- ◆ Ice crystals fracture the cells, creating a hypertonic tissue environment

- ◆ Chemical burns

- ◆ Caused by a wide range of caustic reactions

- ◆ In addition to the duration of exposure, the *chemical nature of the agent* will determine the injury severity

Types of Skin Injuries (cont.)

◆ Electrical current (6%)

- ◆ Electrical energy is transformed into thermal energy as the current passes through poorly conducting body tissues, disrupts membrane potential and function
- ◆ Magnitude of injury directly correlates to strength and duration of current, current pathway, and resistance of tissues

◆ Inhalation

- ◆ Injury of airway tissues that frequently occur with flash burns from fire or steam
- ◆ Smoke burns the pharynx // steam can burn the airway below the glottis // toxic chemicals can produce chemical burns in the lower airway

◆ Radiation

- ◆ Most common: sunburn!!!
- ◆ Radiation therapy can also cause burns

review: SKIN FUNCTIONS

- Barrier against infection
- Temperature regulation
- Body fluid regulation
- Excretion of sweat, electrolytes
- Synthesis of Vitamin D
- Cosmesis

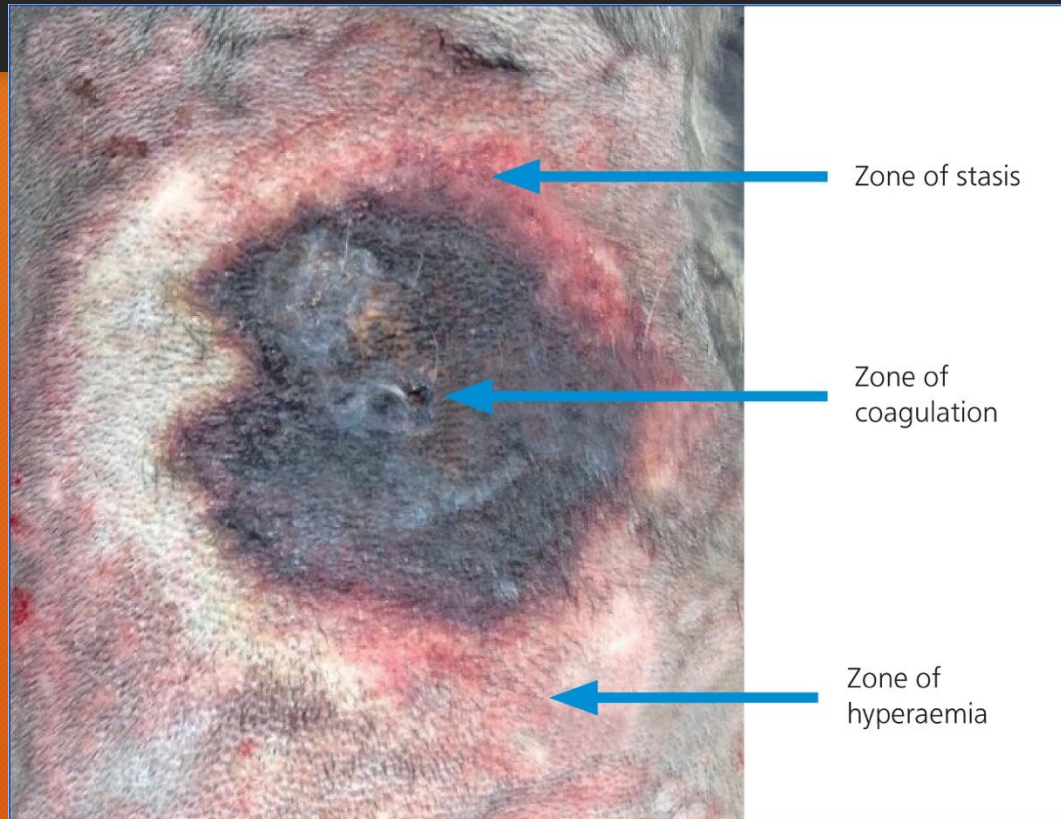
Determinants of Burn injury

- Temperature intensity
- Duration of contact
- Source
 - Flame, liquid, chemical, electrical

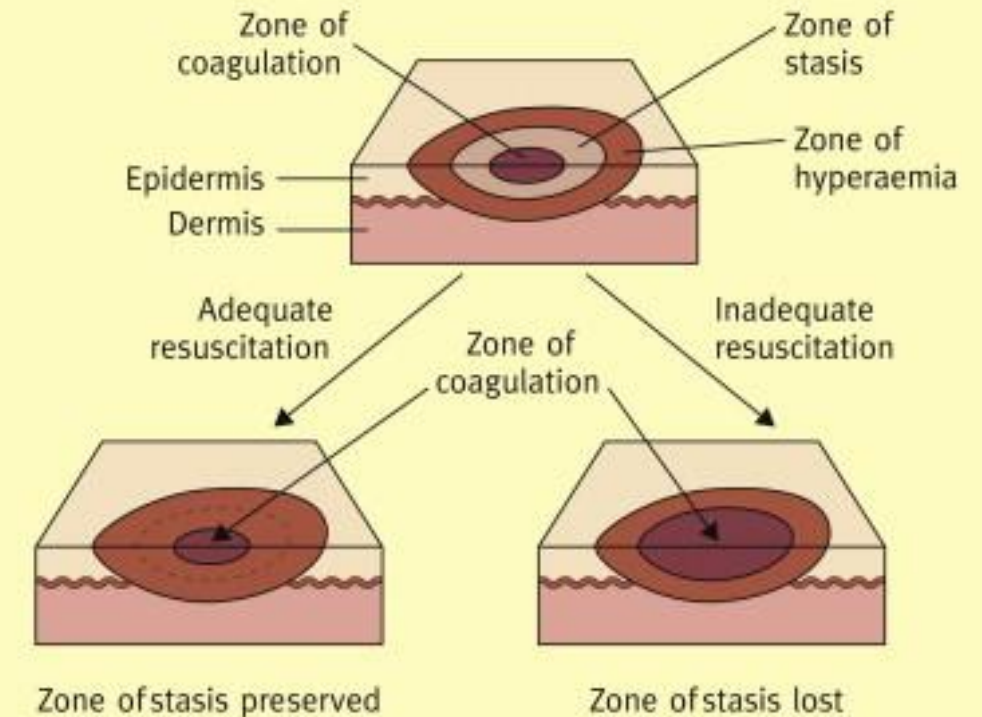
What happens in burn injury?

- Local response (at the site of the burn)
 - Direct effect of heat → cellular proteins coagulate → tissue damage and ischemia → the subsequent reaction to the burn
 - Jackson Model of Burns
- Systemic response (how the entire body could respond)
 - Loss of water, electrolytes, etc. → increased vascular permeability and hypovolemia
 - In severe burns, the body can enter a hypermetabolic state resulting in muscle wasting, tachycardia, hypertension, and other features
 - Immunosuppression and impaired skin barrier function, susceptible to certain bacterial infection (i.e. Pseudomonas)
 - Systemic Inflammatory Response, leading to Acute Respiratory Distress Syndrome (ARDS)
 - Curling Ulcers (gastric erosion ulcers) can appear in the setting of severe burns!

Zones of burn injury: Jackson Model



Jackson's burn zones



Reference:

<https://basicsofburncare.org/pathophysiology/>

Classification of Burns: New vs. Old Nomenclature

Old Terminology	Revised Terminology
1 st degree	Superficial a.k.a. epidermal
2 nd degree	Partial-thickness (superficial, deep)
3 rd degree	Full-thickness (the worst burns can extend in SQ tissue and involve fascia, muscle, nerve, or bone)
4 th degree	

Surgical intervention is typically needed for any deep-partial thickness burns or worse!

Depth of burn

Level

Superficial

Superficial partial thickness

Deep partial thickness

Full thickness

Deep thermal necrosis

Epidermis

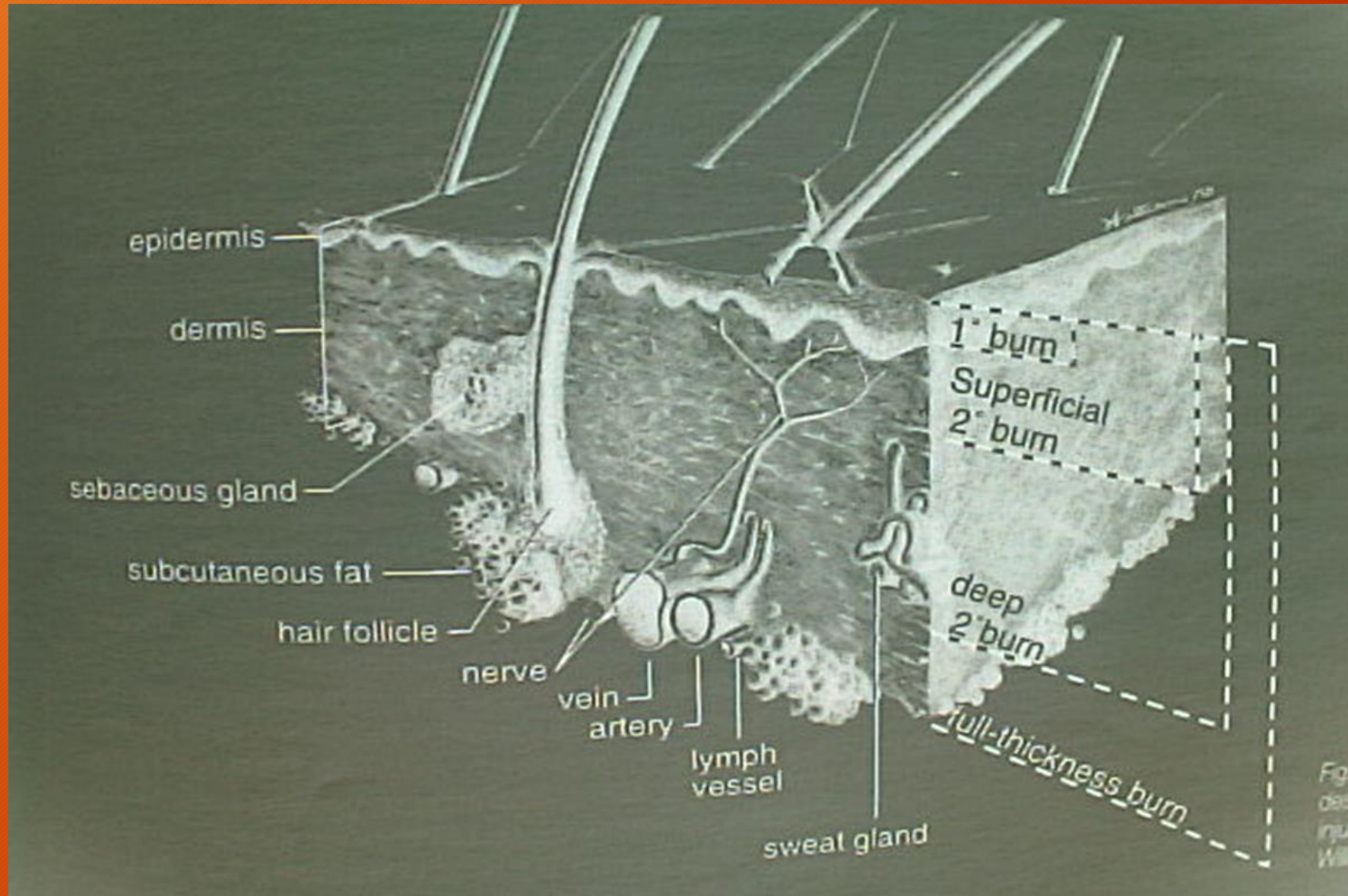
Dermis

Subcutaneous tissue

Muscle



Burn Depth Assessment



Superficial burns

Involve *only* the epidermal layer of skin

- ◆ Classic sunburn
- ◆ Do *not* blister
- ◆ Painful, dry, red, and blanch with pressure
- ◆ Pain and erythema subside in 2-3d
- ◆ By day 4: injured epithelium peels away from the newly healed epidermis
- ◆ Generally healed within 2-6 days *without* scarring
- ◆ No grafting, intervention needed



Superficial Partial-thickness

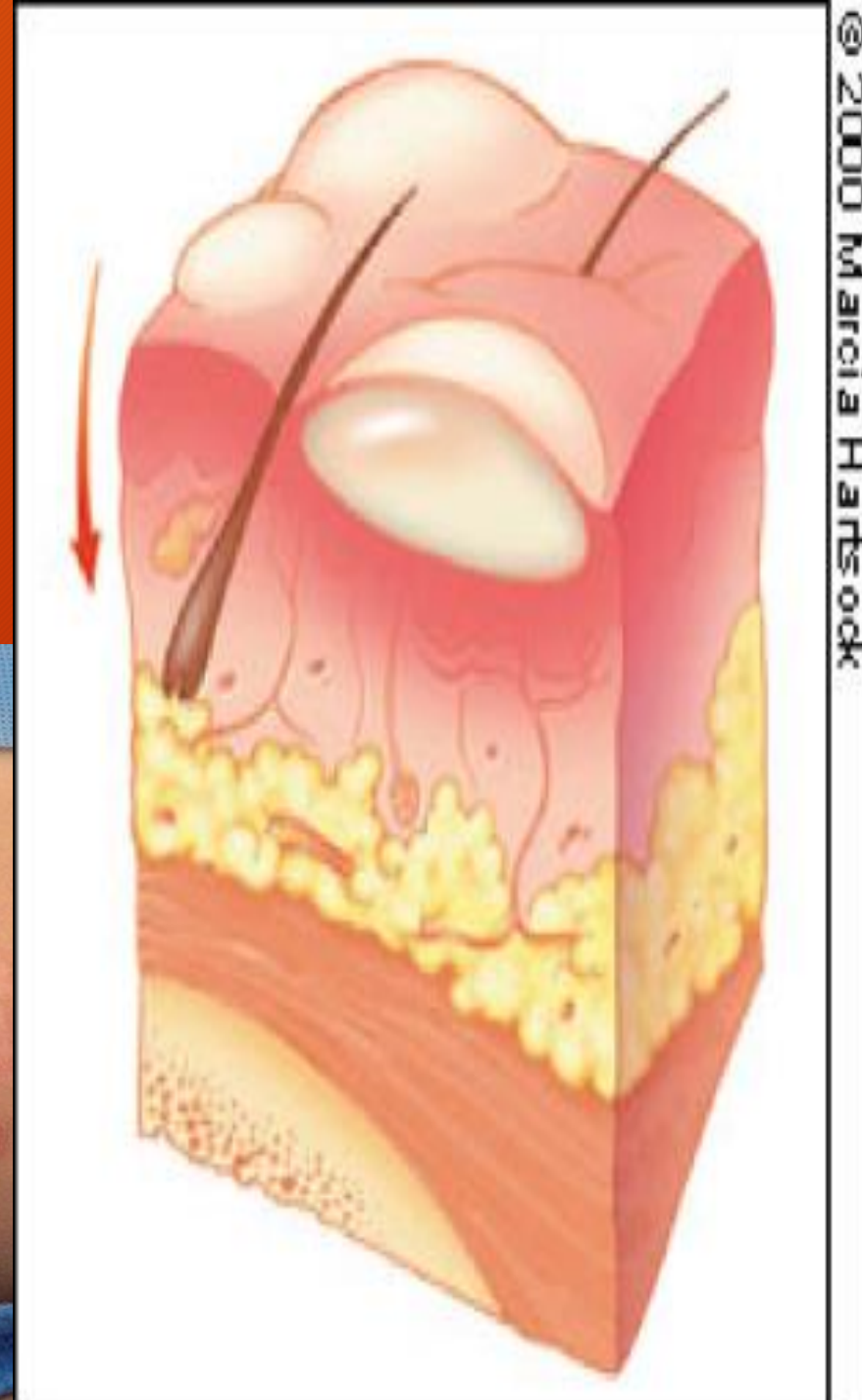
Between the epidermis and dermis

- Form *blisters* within 24h between the epidermis and the dermis;
- Epidermis has been destroyed; Melanocytes destroyed
- Blisters, edema; **very painful**, red, weeping, and blanch with pressure → even movement of air along the wound edges can trigger pain
- Initially may appear to be only epidermal in depth and may be determined to be partial-thickness 12-24 hours later (burn demarcation)
- Generally heal in 7 to 21 days
- Scarring is unusual, although temporary pigment changes may occur
- No grafting, intervention needed

Image source: “[Scalded skin with thin-walled blister. Second degree burn after 2 days.](#)” by Themidget17, [Wikimedia Commons](#), licensed under [CC BY-SA 4.0](#).



Superficial
Partial-
thickness



Superficial Partial-thickness (cont.)

Between the epidermis and dermis

- ◆ Fibrinous exudates and necrotic debris may accumulate on the surface
- ◆ Predisposes the burn to heavy bacterial colonization, delayed healing
- ◆ Typically heal without functional impairment or hypertrophic scarring



Deep Partial-thickness

Extends into the deeper dermis



- *Damaged hair follicles, nerve ending, and sweat glands*
- Painful to pressure only
- Almost always blister (easily unroofed)
- Appear wet or waxy, dry, red/tan/white coloration
- Mottled colorization from patchy cheesy white to red
- Do NOT blanch with pressure
- Healing time 21-35 days IF no infection; *some* regeneration only
- IF CIRCUMFERENTIAL: will require ESCHAROTOMIES

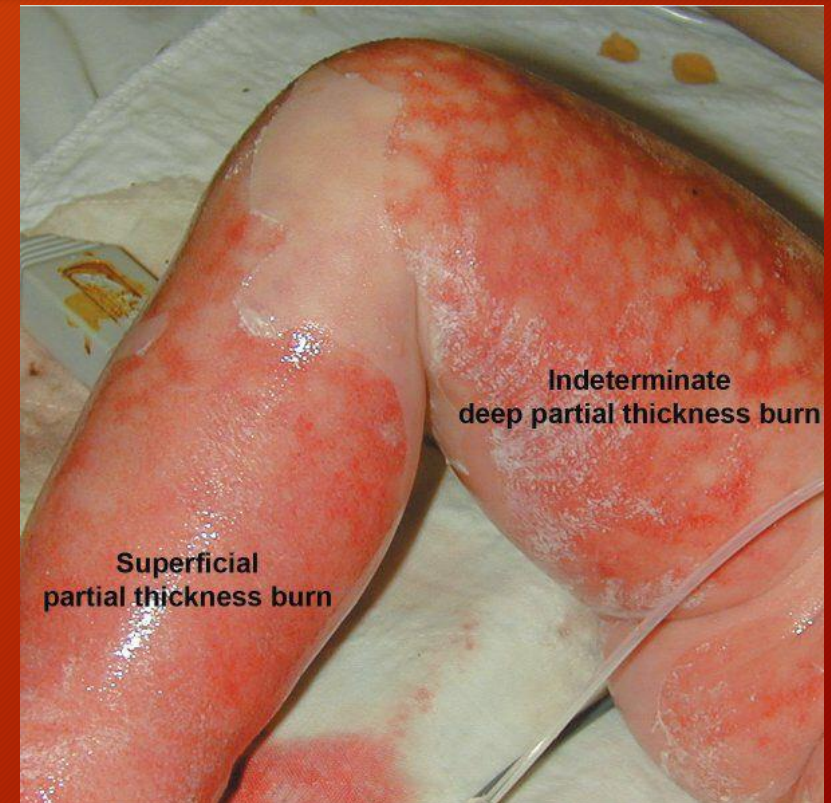
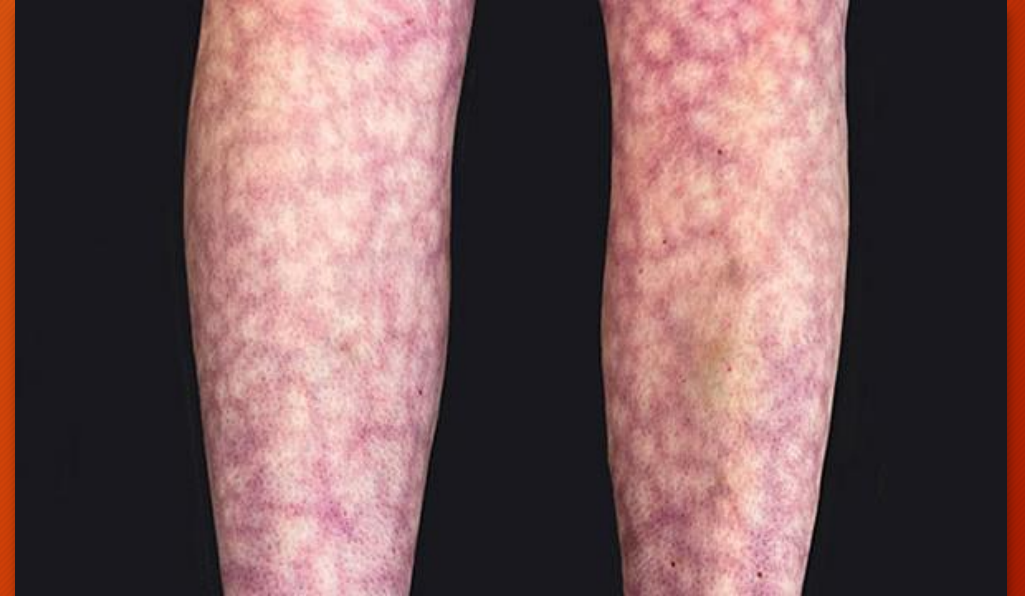
Be on the lookout...for red flags.

- ◆ Admit to the hospital *ANY CHILD* whom you suspect may have been burned intentionally:
 - ◆ Repeat visits to ED
 - ◆ Odd burn patterns (cigarettes, iron, curling iron)
 - ◆ Injury description is incongruous with child's development
 - ◆ Burns in children too young to walk
 - ◆ e.g. "she fell in the bathtub" does not explain a buttock burn in an 6-month old
 - ◆ "Pt tripped over herself"...but if the pt is not yet able to walk or crawl...

Derm terminology: Mottled skin

Occurs when blood flow to tiny vessels under your skin is disrupted. This results in a fine, bluish-red, lace-like pattern (reticula). The condition is also known as livedo reticularis.

Deep partial-thickness burns show mottled skin with red and/or white patches.



Deep Partial-thickness

Extends into the deeper dermis

- ◆ All epidermal/dermal layers destroyed
- ◆ Leathery, waxy/pearly; cause hypertrophic scarring
- ◆ If a joint is involved, joint dysfunction is expected
- ◆ If failure to heal in 3 weeks, are functionally and cosmetically equivalent to a full-thickness burn
- ◆ Insensate; differentiation from a full-thickness burn is often difficult
- ◆ Heal in 3-9 weeks if no grafting needed



Full-thickness

Extends through and destroy all layers of the dermis...and often injury to the underlying subcutaneous tissue

- ◆ Skin appears waxy white-to-leathery gray to charred and black
- ◆ Dry, inelastic, does NOT blanch with pressure
- ◆ Burn eschar (dead and denatured dermis) is usually intact
- ◆ Eschar can compromise the viability of a limb or torso if circumferential
- ◆ Hairs can easily be pulled from follicles
- ◆ Vesicles and blisters do not develop



Full-thickness

- ◆ Pale full-thickness may simulate normal skin except the skin does NOT blanch
- ◆ Eschar eventually separates from underlying tissue and reveals an unhealed bed of granulation tissue
- ◆ Without surgery, heal by wound contracture with epithelialization around wound edges
- ◆ Scarring is severe with contractures
- ◆ Complete spontaneous healing is not possible





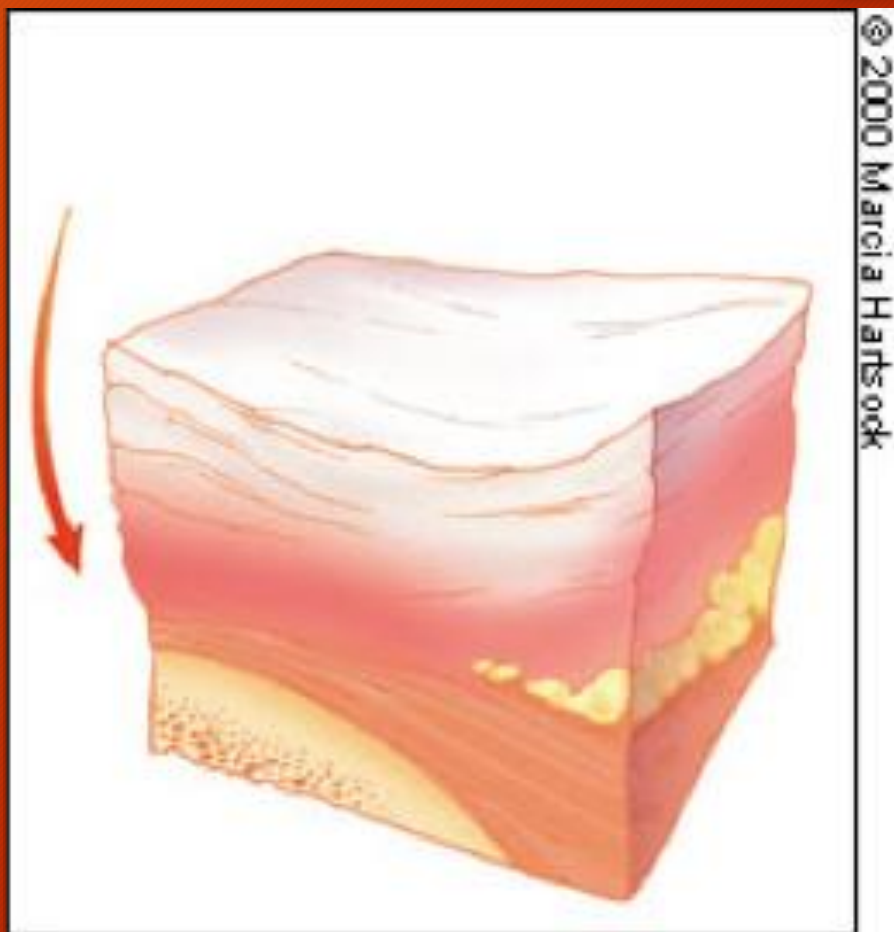
Full-thickness burn before tx (TBSA 3%)



After skin graft + biodegradable temporizing matrix (BTM)

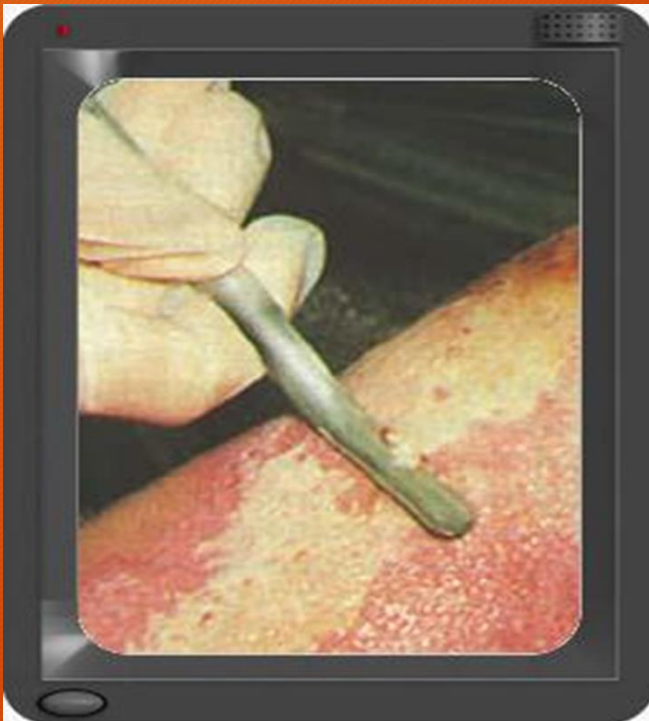
Image reference:

https://www.researchgate.net/publication/374228571_The_use_of_NovoSorb_Biodegradable_Temporising_Matrix_BT_M_in_the_reconstruction_of_complex_soft_tissue_defects_-_an_oncological_aesthetic_and_practical_solution



Fourth Degree Burns

- ◆ Deep and potentially *life-threatening injuries* that extend through the skin into underlying tissues such as fascia, muscle, and/or bone



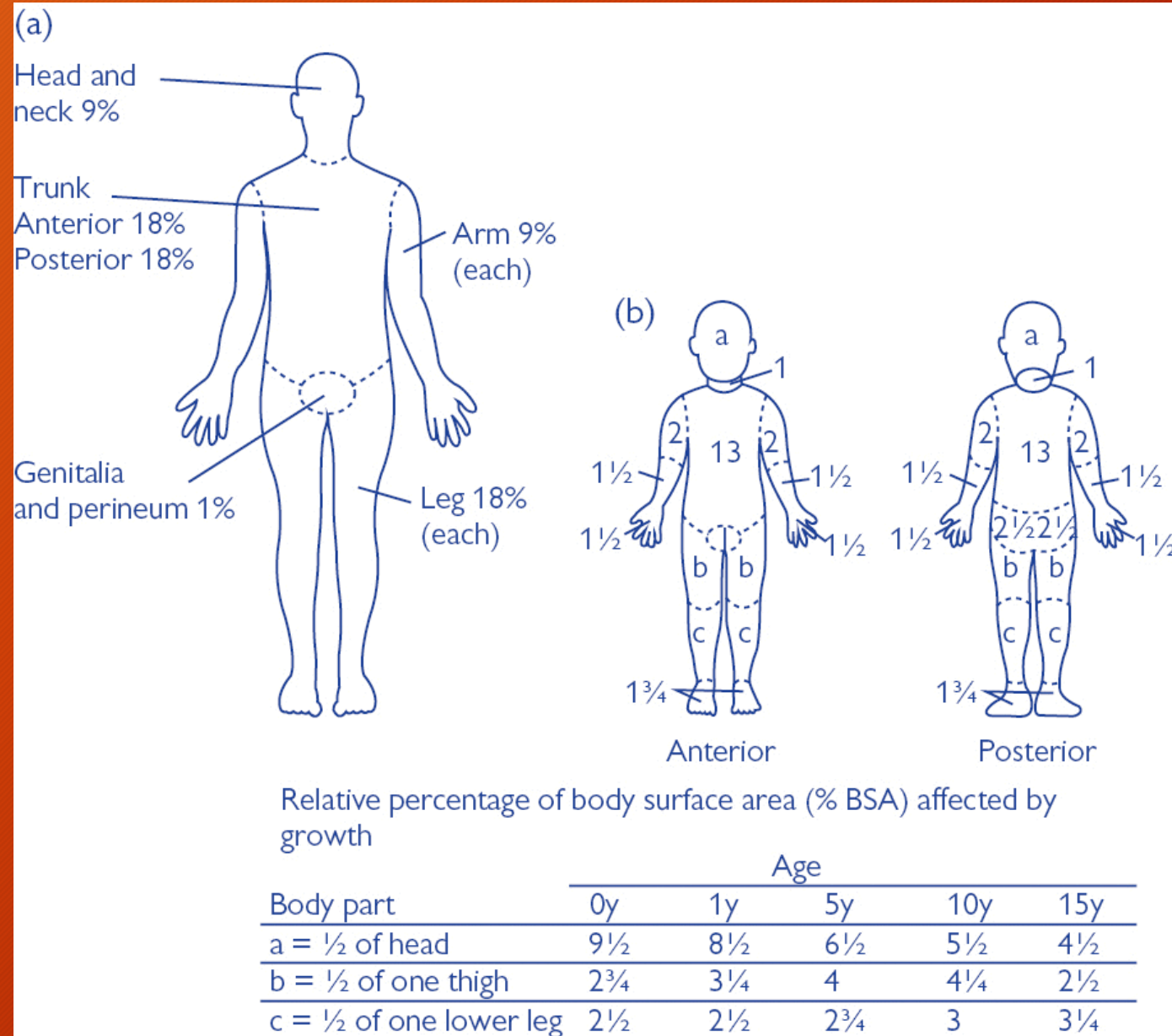
New system (old system)	Superficial (first degree)	Superficial partial- thickness (second degree)	Deep partial-thickness (second degree)	Full-thickness (third degree)	Full-thickness (fourth degree)
Etiology	Ultraviolet light, very short flash (flame exposure)	Scald (spill or splash), short flash	Scald (spill), flame, oil, grease	Scald (immersion), flame, steam, oil, grease, chemical, high-voltage electricity	Cause as for deep partial-thickness burns
Histology	Epidermis only	Epidermis and papillary dermis, skin appendages intact	Epidermis and reticular dermis, most skin appendages destroyed	Epidermis and dermis; all skin appendages destroyed	Involves fascia and muscle and/or bone
Clinical features (appearance)	Erythema, dry and pink or red; blanches with pressure	Erythema, blisters; moist, red and weeping; blanches with pressure	Blisters (easily unroofed); wet or waxy dry; variable color (patchy to cheesy white to red); does not blanch with pressure	Waxy white to leathery gray to charred and black; dry and inelastic; does not blanch with pressure	Black (dry, dull, and charred)
Sensation	Painful	Painful to air and temperature	Perceptive of pressure only	Deep pressure only (insensate)	Deep pressure only (insensate)
Healing time	3 to 6 days	7 to 20 days	More than 21 days	Never (no spontaneous healing)	Never (no spontaneous healing)
Scarring	None	Unusual; potential pigmentary changes, moist, elastic	Severe (hypertrophic) risk of contracture	Very severe risk of contracture, eschar tissue (hard and inelastic)	Eschar tissue (hard and inelastic)
Treatment	Cleaning, cooling with running water or a cold compress, return of full function	Cleaning; cooling with running water or a cold compress, topical agent; sterile dressing, the return of full function	Cleaning; topical agent; sterile dressing; possible surgical excision and grafting; earlier return of function with surgery	As for deep partial-thickness burns plus surgical excision and grafting at the earliest possible time, functional limitation more common if not grafted	Healing requires surgical intervention, functional impairment
Complication	Increase the risk of skin cancer	Local infection, cellulites	Possible skin grafting	Possible amputation	Amputation, gangrene, death

Burn Size estimation = a Crucial Skill

- ◆ Often underestimated
- ◆ Expressed as Total percentage of Body Surface Area (TBSA)
 - ◆ Superficial burns (1st degree) are not included in calculation
- ◆ Several ways of estimating burn size:
 - Lund-Browder: most accurate assessment but inefficient
 - “Rule of Nines” = great for rapid assessment, efficient but can over-estimate
 - Palm method = good for minor burns, least accurate
- ◆ Goal of calculating TBSA: determine and/or your treatment!

Lund-Browder chart

- ◆ Most accurate method for estimating TBSA for both adults and children
- ◆ Takes into account the relative percentage of BSA affected by growth



Wallace's Rules of Nines

- Assesses the proportion of body surface area for partial and/or full-thickness burns
- Provides efficient assessment for pts between 10-80 kg
 - Estimates are less accurate for pediatric and obese patients
 - i.e. Obese patients have torsos whose percentage of TBSA is greater
- Risk of overcalculation → excessive fluid resuscitation

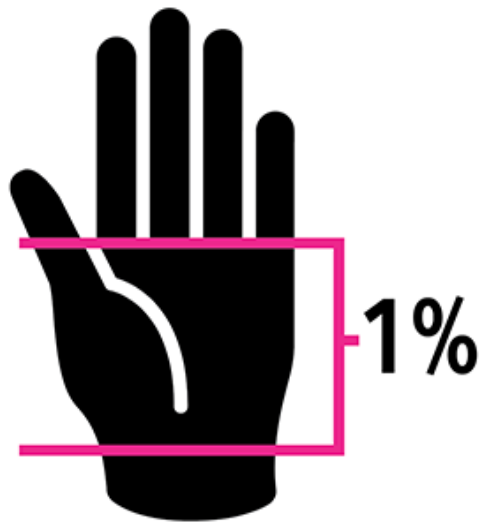
Rule-of-9s	Total	Subdivision
Head	9%	Anterior Head = 4.5% Posterior Head = 4.5%
Torso	18%	Chest = 9% Abdomen = 9%
Back	18%	–
Each Arm	9%	Anterior Arm = 4.5% Posterior Arm = 4.5%
Each Leg	18%	Anterior Leg = 9% Posterior Leg = 9%
Perineum	1%	–



Alterations to the Rule of Nines

- ◆ For patients with BMI > 30, The "Rule of Fives" has been proposed to overcome the inaccuracies of estimating TBSA in obese patients, given that more than 50% of TBSA is contributed by the trunk.
 - ◆ This formula assigns each arm 5% TBSA, each leg 20%, and 50% for the trunk.
- ◆ For patients weighing less than 10kg (pediatric), A "Rule of Eights" has been proposed as a better TBSA estimate for patients weighing less than 10 kg.
 - ◆ This rule dictates approximately 32% TBSA for the patient's trunk, 20% TBSA for the head, 16% TBSA for each leg, and 8% TBSA for each arm.

Palm Method = if you're in a pinch



ESTIMATE BURN SIZE

Use the patient's palm to measure the body surface area burned. The palm of the patient - not fingers or wrist area, is about 1% of the BSA (Body Surface Area).

◆ NOTE: Small or patchy burns can be approximated by using the patient's palm

Burn patients should be transferred asap *if*:

- 10% or more in ages <10 or >50 yrs of age
- TBSA burn 20% or more, any ages → Burn shock
- Burns to face, hands, feet, or perineum
- Chemical, electrical, inhalational, multiple comorbidities, or combined trauma + burn

Burn Complications

- Burn Shock: increased capillary permeability and myocardial depression → large fluid shifts resulting in tissue edema and vascular volume depletion
 - Occurs within the first 1-2 days s/p severe burn
 - Rapid resuscitation protocols prevent this!
- Changes in metabolic response: hypometabolism within the first 2 days → hypermetabolism gradually sets in after that
 - DIRECTLY CORRELATED WITH MAGNITUDE OF BURNS
 - Therefore, important to also consider nutritional support as part of tx regimen

Fluid Resuscitation - Parkland formula

- Represents the amount (in mL or L) of fluid to be given over 24 hours
 - for partial-thickness burns or worse
- $[4 \text{ mL Lactated Ringer's} \times \% \text{ TBSA} \times \text{weight (kg)}] / 24 \text{ hours}$
- Half of total fluid should be given in first 8 hours, 2nd half should be given over the next 16 hours.
 - Therefore, your hourly fluid rate in the first 8 hours is double that of the next 16 hours
- After the first 24 hours, proper fluid hydration can be determined via monitoring urine output.

Fluid Resuscitation - Parkland formula

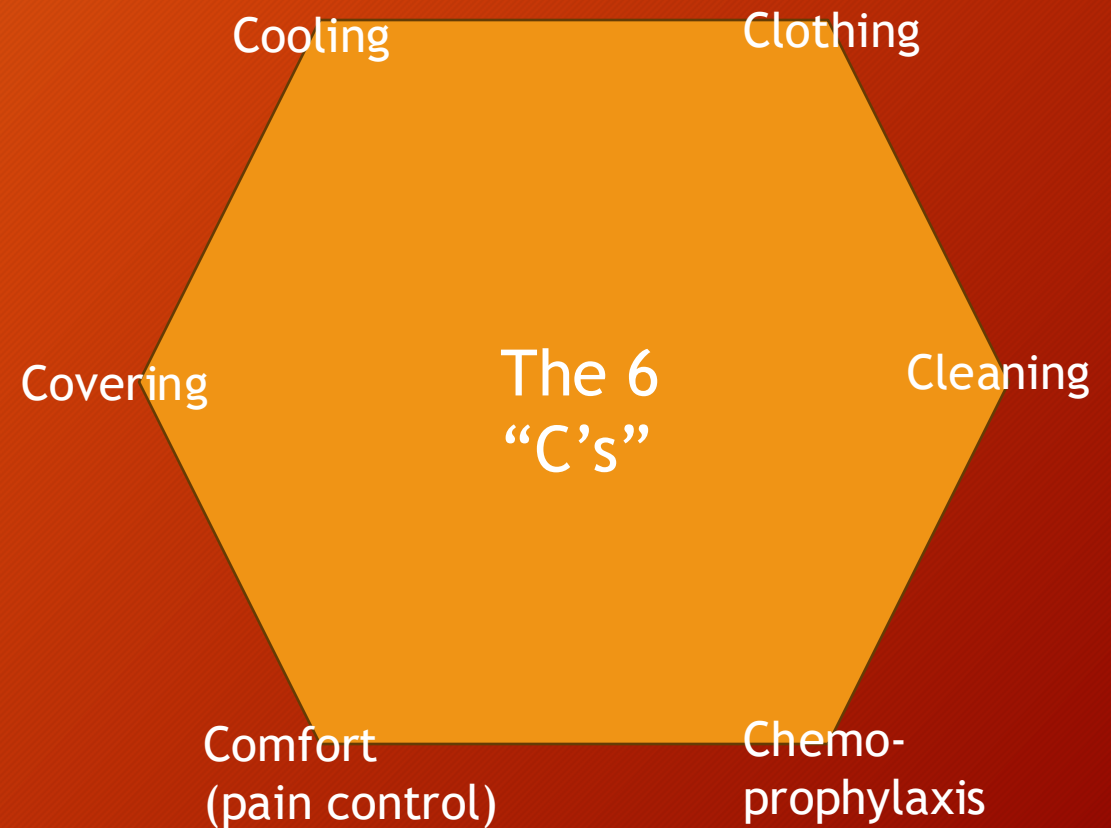
- The goal of proper fluid hydration is to prevent vital organ damage
- Excessive resuscitation can result in pulmonary edema (congestion), possible ARDS, or even compartment syndrome

Treatment of Superficial Burns or Limited Partial-thickness Burns

- ◆ Outpatient management with "the 6 C's"

- ◆ Clothing
- ◆ Cooling
- ◆ Cleaning
- ◆ Chemoprophylaxis
- ◆ Covering
- ◆ Comforting (pain control)

- ◆ And: always at least *consider* referral (to burn center or plastics)! Be preemptive



Treatment of Superficial Burns or Limited Partial-thickness Burns

- ◆ Clothing: remove hot or burned clothing
 - ◆ Clothes may be melted to skin, so be careful when removing them!!!
- ◆ Cooling: cool region (to approx. 55°F) for 10-30 min
 - ◆ Under a faucet or with compress
 - ◆ No ice packs!
 - ◆ May cause frostbite and further damage skin

Treatment of Superficial Burns or Limited Partial-thickness Burns

◆ Cleansing:

- ◆ Wash gently with mild alcohol-free soap, then normal saline daily
- ◆ Blot dry - do not rub
- ◆ Embedded materials should be removed by copious irrigation using a large-gauge syringe

Treatment of Superficial Burns or Limited Partial-thickness Burns

◆ Chemoprophylaxis:

- ◆ Tetanus immunization (partial-thickness burns)
- ◆ Routine skin cultures are NOT recommended except when infection is suspected
- ◆ Prophylactic systemic antibiotics are NOT recommended
- ◆ Partial-Thickness burns are sometimes treated with a topical antimicrobial agents such as silver sulfadiazine, bacitracin, or silver-impregnated synthetic dressings

Treatment of Superficial Burns or Limited Partial-thickness Burns

◆ Covering:

- ◆ All partial-thickness burns should be covered with sterile dressings

◆ Comfort:

- ◆ Analgesics around-the-clock (ATC) are recommended
- ◆ Tylenol and NSAIDs alone or in combo with opioids
- ◆ “rescue” analgesics before dressing changes and physical activity

Dressing Type	Main Uses	Absorption	Contraindications
Plastic films	• Epithelialization of wounds • Support autolysis • Protection of new epithelial tissue	Δ	• Draining wounds • Infected wounds • Poorly granulated wounds
Hydrogels amorphous wafer	• Hydration of dry wounds • Support autolysis • Donor sites (grafts) • Epithelialization	+	• Excessively draining wounds • Infected wounds (wafer)
Hydrocolloids	• Granulation tissue formation • Support autolysis	+	• Infected wounds • Excessively draining wounds
Calcium alginates	• Absorption of exudate • Hemostasis • Infected wounds	++	• Superficial wounds • Epithelializing wounds
Hydrofibers	• Absorption of exudate into the fibers (dynamic) • Infected wounds	+++	• Epithelializing wounds
Foams adhesive nonadhesive	• Excessively draining wounds • Nondynamic absorption	+++	• Infected wounds (adhesive)
Biological agents	• Difficult cases • Dormant wounds	Δ	• Infected wounds • Necrotic wounds

$\Delta = 0$

+ = *small amount*

++ = *moderate amount*

+++ = *large amount*

Treatment of Superficial Burns or Limited Partial-thickness Burns

◆ Follow up:

- ◆ See pt. back in 24 hours, then again 48-72 hours
- ◆ Follow up weekly until wound epithelialization occurs
- ◆ Refer to burn specialist if epithelialization hasn't occurred in 2 weeks
- ◆ ...there are at risk for contracture



ABA Admission Outpatient Guidelines

- ◆ May treat as outpatient if:
 - ◆ <10% TBSA burn in adult
 - ◆ <5% TBSA in child or elderly
 - ◆ <2% full-thickness burn
- ◆ Hospital admission or burn center referral if:
 - ◆ High voltage injury
 - ◆ Suspected inhalation injury
 - ◆ Circumferential burn
 - ◆ High risk patient (DM)

Mortality Risk

- ◆ Major risk factors

- ◆ Age >60y OR <5y

- ◆ >40% TBSA burned

- ◆ Inhalation injury

- ◆ Mortality:

- ◆ Zero risk factors: 0.3%

- ◆ 1 risk factor: 3.0%

- ◆ 2 risk factors: 33%

- ◆ 3 risk factors: 90%

Surgical management of Burns

- Skin Grafts
 - Autograft: patient's own skin; no rejection
 - Allograft: cadaveric; 2-4 wks coverage before rejection
 - Biobrane: bovine collagen and nylon; 2nd degree burns
 - Integra: permanent; bovine collagen; autograft also
 - Cultured Epidermal Autograft (CEA): grown in lab from patient's own skin; take 17-25 days

Factors in Graft Healing

- Infection
- Poor debridement
- Poor quality donor site skin
- Poor nutrition
- Bleeding/hematoma under graft
- Mechanical/sheering trauma to graft
- Shock
- Grafting over joint, bony area
- ROM to graft area with disruption (too early)

Inhalation Injury

- ◆ Those with suspected inhalation injury should be admitted for observation for 12-24 hours
 - ◆ Can quickly decompensate
- ◆ Vigilant: coughing, wheezing, dyspnea, facial burns, sooty mucus; burned eyelashes or facial/nasal hair
- ◆ Monitor carbon monoxide (CO) levels with ABGs
- ◆ Consider bronchoscopy and/or preemptive intubation

Inhalational injury

Inhalational Injury



- Airway burns and swelling
- Direct lung injury
- May take hours to swell
- Singed facial hair is worrisome

Signs of Inhalation Injury

Also look
for
soot/dirt
around
the
nostrils



- ▣ Burns of face or mouth
- ▣ Singed facial hair
- ▣ Sooty sputum
- ▣ Hoarse voice or stridor
- ▣ Cough or dyspnea
- ▣ History of exposure in enclosed space

Pulmonary complications of inhalational injury

➤ Smoke Inhalation

- Carbon monoxide, other chemicals



i.e. Keep track of pt's oxygenation + LOC
- Order Carboxyhemoglobin levels

➤ Steam Inhalation

- Direct thermal injury to upper airway

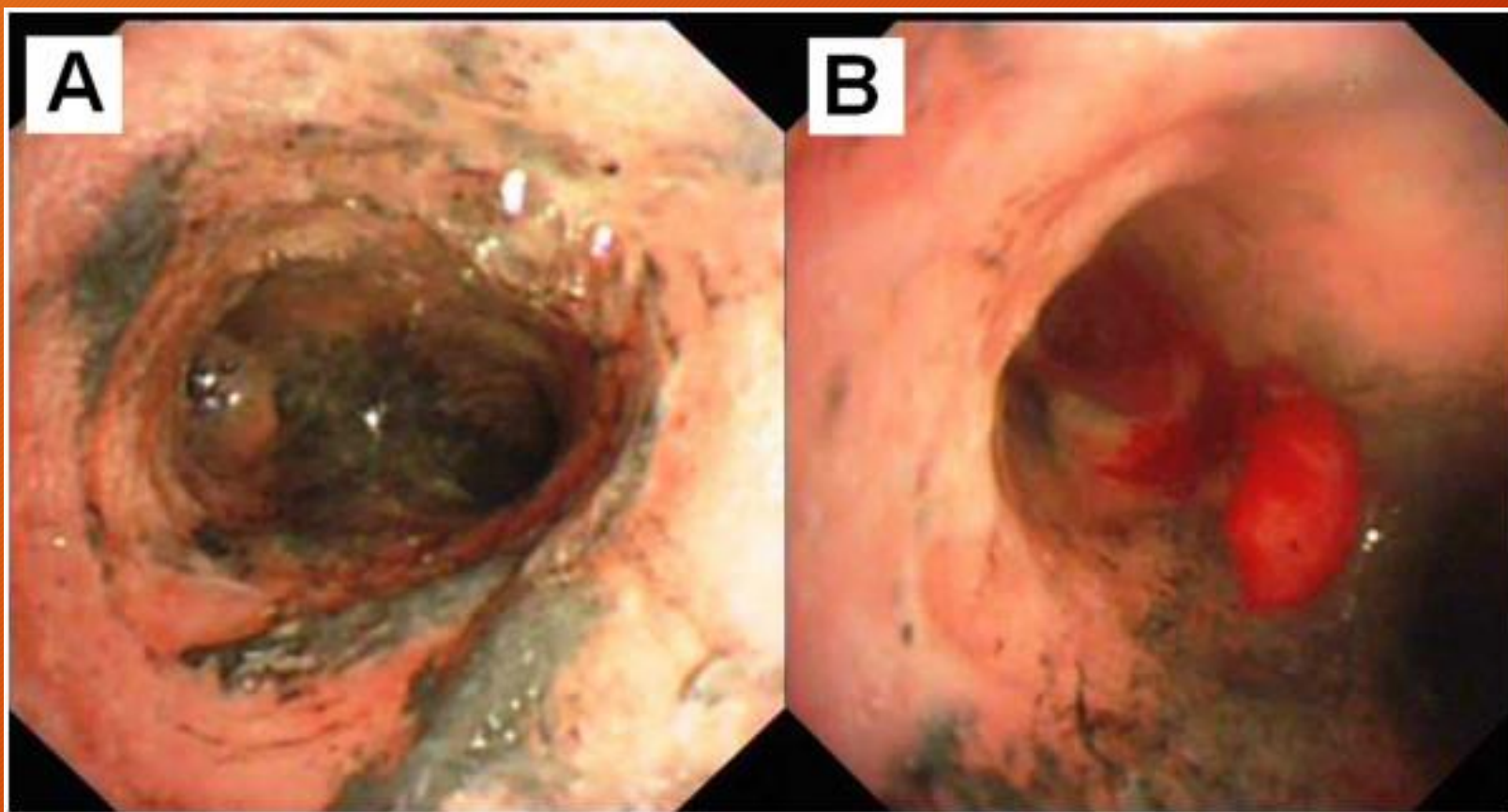


i.e. Keep track of pt's oxygenation + RR + work of breathing

➤ Chest Wall Restriction

- Burn injury with decrease wall motion, decreased inspiratory ability





Chronic v. Acute wound care

Some basic terminology

Acute wound: a disruption of skin and/or underlying soft tissue that *typically has a well-organized healing process with predictable tissue repair*

- examples: bruises, lacerations, abrasions, burns

Chronic wound: *a wound with an impaired healing process*, usually involving a prolonged or excessive inflammatory phase

- Typically cells cannot respond as effectively to reparative stimuli
- ALL CHRONIC WOUNDS BEGIN AS ACUTE WOUNDS
- Examples: diabetic ulcers, pressure ulcers (decubitus ulcers), vascular ulcers

“Open” vs. “closed”: relates to skin breakage

- An open wound is one with skin breakage and the underlying tissue is exposed to the outside environment (i.e. GSWs, punctures, lacerations)
- A closed wound is one with intact skin (i.e. crush injuries, hematomas)
- This differentiation is also used when discussing fractures

Principles of Wound Healing

1. For proper healing, the wound bed needs the following:

- Sufficient vascularization (blood flow!)
- To be free of devitalized tissue:
 - After removal of this tissue, you say that “the wound is debrided”
- To be without infection
- Moist

2. And therefore, wound dressings should eliminate dead space, control exudate, prevent bacterial overgrowth, and ensure proper fluid balance

Medical therapy Considerations

◆ Antibiotics

- ◆ All wounds are colonized with microbes but not all wounds are infected
- ◆ No evidence to support prophylaxis
- ◆ Look for clinical signs for indications to treat with abx
 - ◆ Local: cellulitis, lymphangitic streaking, purulence, malodor, wet gangrene, osteomyelitis
 - ◆ Systemic: fever, chills, nausea, hypotension, hyperglycemia, leukocytosis, confusion
- ◆ Target blood glucose control (80-110) --> high blood sugars can delay wound healing (interferes with WBCs) and leave patient prone to localized infection
 - ◆ signs of uncontrolled hyperglycemia might indicate an underlying chronic issue (i.e. T2DM)

Nutritional Support for Wound Care

- ◆ Key nutrients for wound healing: according to [Memorial Sloan Kettering Cancer Center](#)
 - ◆ Protein - helps body build tissue (i.e. chicken, beef, lentils, peas, quinoa, eggs, fish, nuts)
 - ◆ Magnesium - helps heal broken tissue and reduce swelling (i.e. avocado, almonds, leafy greens, banana, whole grains)
 - ◆ Zinc - helps skin heal and grow (i.e. seafood, meats, yogurt, spinach, eggs, nuts and seeds)
 - ◆ Vitamin A - helps body make more skin cells (i.e. sweet potatoes, broccoli, carrots, mango, melons, milk, tomatoes, leafy greens)
- ◆ “The presence of a wound may increase metabolic demands due to the metabolic response to injury. In addition, large quantities of protein can be lost in wound exudates each day. It is not uncommon to see large pressure ulcers or large leg wounds that represent 5-10% body surface area. The presence of a chronic wound can increase protein requirements by 250% and calorie requirements by 50% in order to maintain an appropriate LBM. All of these protein losses should be replaced in order to prevent further PCM and complications to wound healing.”¹

Note: There is limited evidence for using serum albumin and prealbumin as proxies for protein deficiencies and malnutrition.²

References:

1. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4217039/>
2. <https://pubmed.ncbi.nlm.nih.gov/21802178/>

Acute Wound Care: general tips

- ◆ Anesthetize the wound
- ◆ Explore the wound and remove foreign bodies
- ◆ No iodine, providone-iodine or hydrogen peroxide! Impairs healing due to neutrophil damage
- ◆ Irrigate with 60ml sterile saline per cm of wound*
- ◆ Debride non-viable tissue
- ◆ Suture or tape wound closed
- ◆ Sterile dressing
- ◆ Give tetanus booster if:
 - wound dirty or complex and it has been more than 5 years since the last booster
 - wound is clean and simple but it has been more than 10 years since last booster
 - patient doesn't know tetanus immunization history or has a decreased level of consciousness
 - Give Tdap to include coverage for whooping cough if giving booster

Chronic Wound Care

- ◆ Debridement
- ◆ Proper wound dressing
- ◆ Controlling underlying factors
 - ◆ Glycemic control
 - ◆ Revascularization

◆Note: If you need to gather samples from an ulcer to determine if it is infected, place your probe or swab in the deeper portions of the wound. Avoid the superficial or wound edges.

◆In short - get in there.



Ulcers

◆ Definition:

- ◆ Open sore that results from the loss of epidermis and part or all of the dermis
- ◆ Possible etiologies: vascular, neoplastic, hematologic, drug-related, connective tissue disease, neurologic, infectious, physical
- ◆ ***Diagnostic staging determines best treatment->best outcome***
- ◆ ***Must differentiate between Chronic Ulcers and Diabetic***

A courtesy trigger warning

◆ Here we go...

Pyoderma gangrenosum

- ◆ Rapidly developing
- ◆ Painful ulcer
- ◆ Undermined, gangrenous border
- ◆ Lower legs
- ◆ Associated with: ulcerative colitis, granulomatous colitis, RA, myeloproliferative diseases

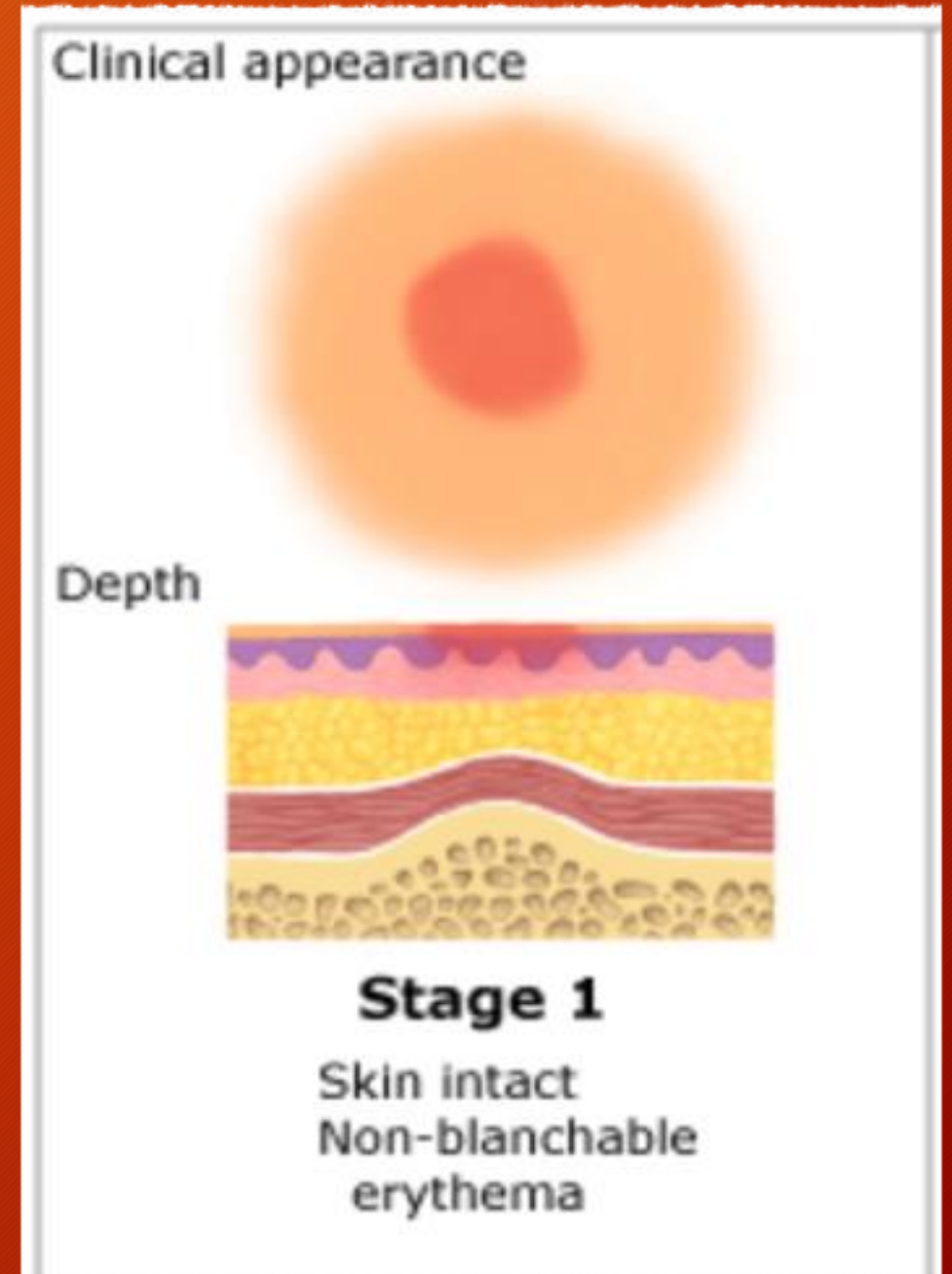


Pressure Ulcers (Decubitus Ulcers)

- ◆ Localized injury to the skin and/or underlying tissue, usually over a bony prominence
 - ◆ i.e. buttocks, heels, elbows
- ◆ Result of pressure
 - ◆ Importance of rotating hospitalized patients and inspecting
- ◆ Superficial skin is less susceptible to pressure-induced damage than *deeper tissue*
- ◆ External appearance may underestimate the extend of pressure related injury

Stage 1 Ulceration

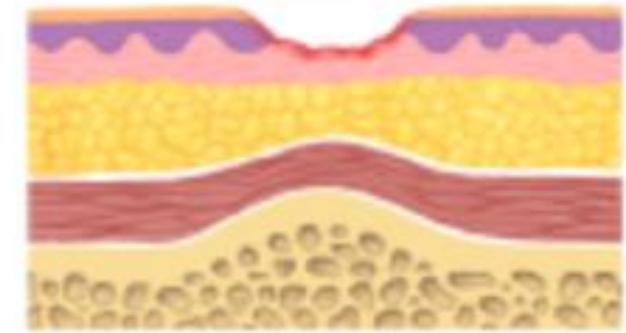
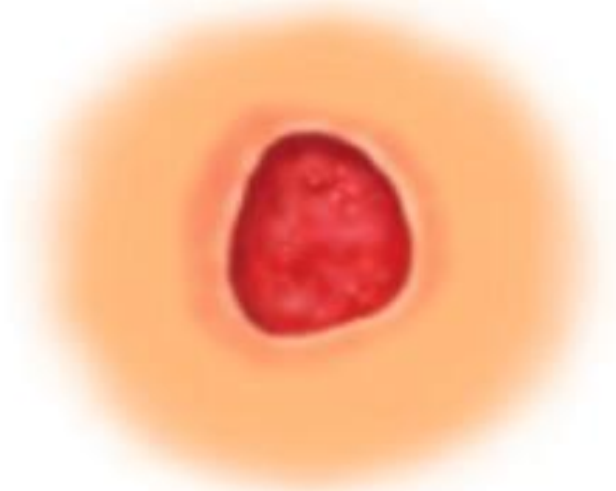
- ◆ Intact skin
- ◆ non-blanchable redness in localized area
- ◆ Usually over a bony prominence





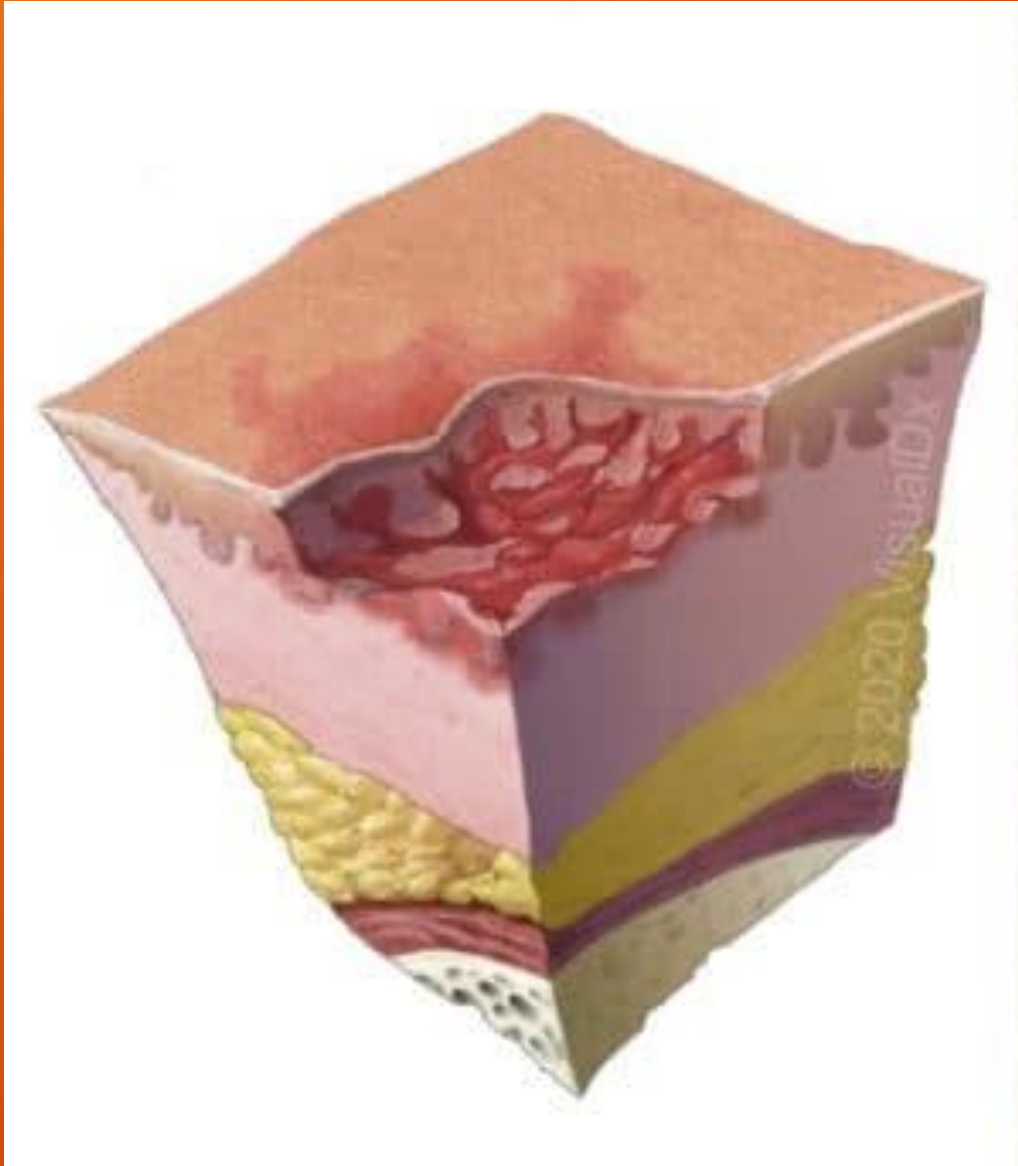
Stage 2 Ulceration

- ◆ Partial-thickness dermal loss
- ◆ Presents as a shiny or dry, shallow open ulcer with a pink-red wound bed
- ◆ Without sloughing or bruising
- ◆ (May be an intact or ruptured serum-filled blister)



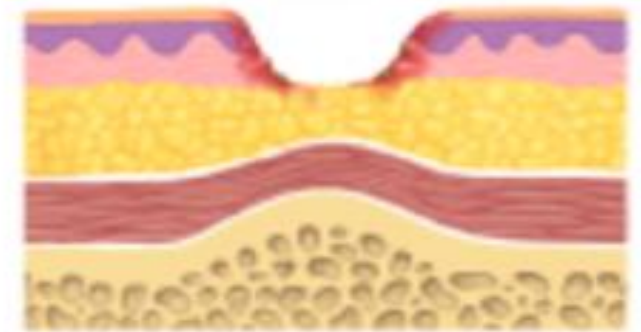
Stage 2

Partial loss of dermis
Shallow open ulcers



Stage 3 Ulceration

- ◆ Full-thickness dermal loss
- ◆ Subcutaneous fat may be visible
- ◆ No exposed bone, tendon, or muscle
- ◆ Undermining and tunneling may be seen



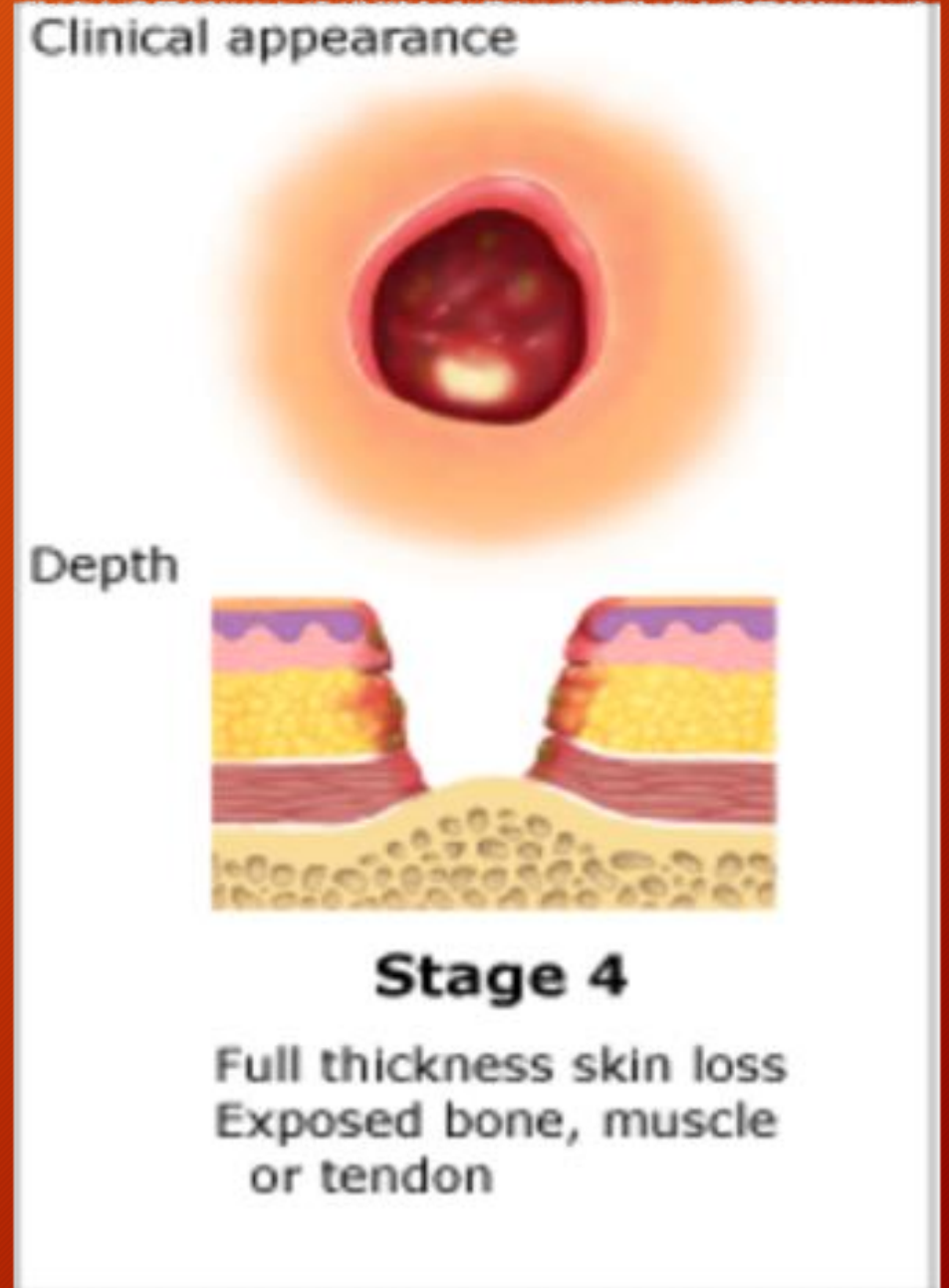
Stage 3

Full thickness skin loss
Fat exposed



Stage 4 Ulceration

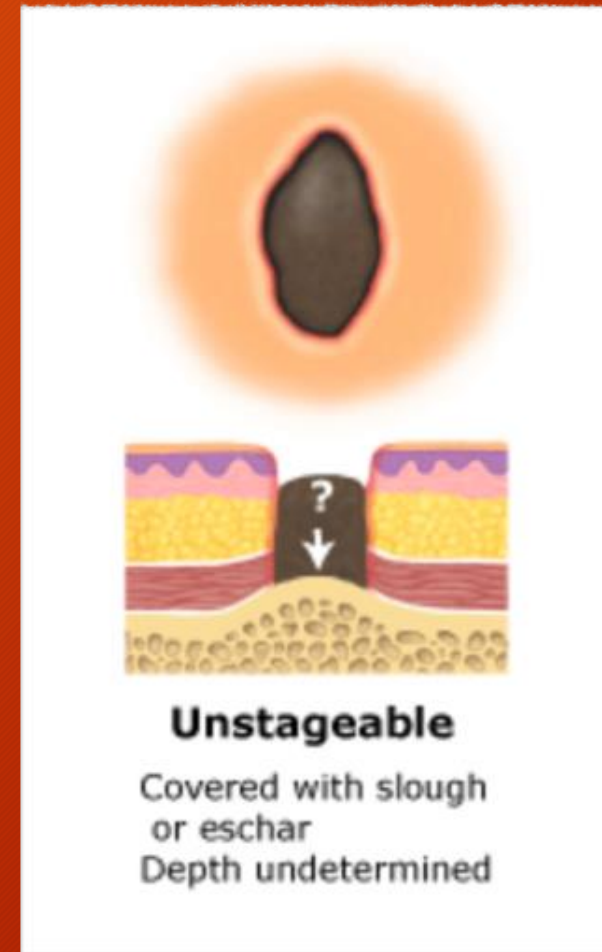
- ◆ Full-thickness skin loss with exposed bone, tendon, or muscle
- ◆ Sloughing or eschar may be present
- ◆ Undermining / fistula formation
- ◆ Possible underlying osteomyelitis





“Unstageable” Ulceration

- ◆ Full-Thickness tissue loss
- ◆ Base of the ulcer is covered by slough and/or eschar
- ◆ Cannot be staged until eschar or slough is debrided





Diabetic Foot Ulcers

Epidemiology

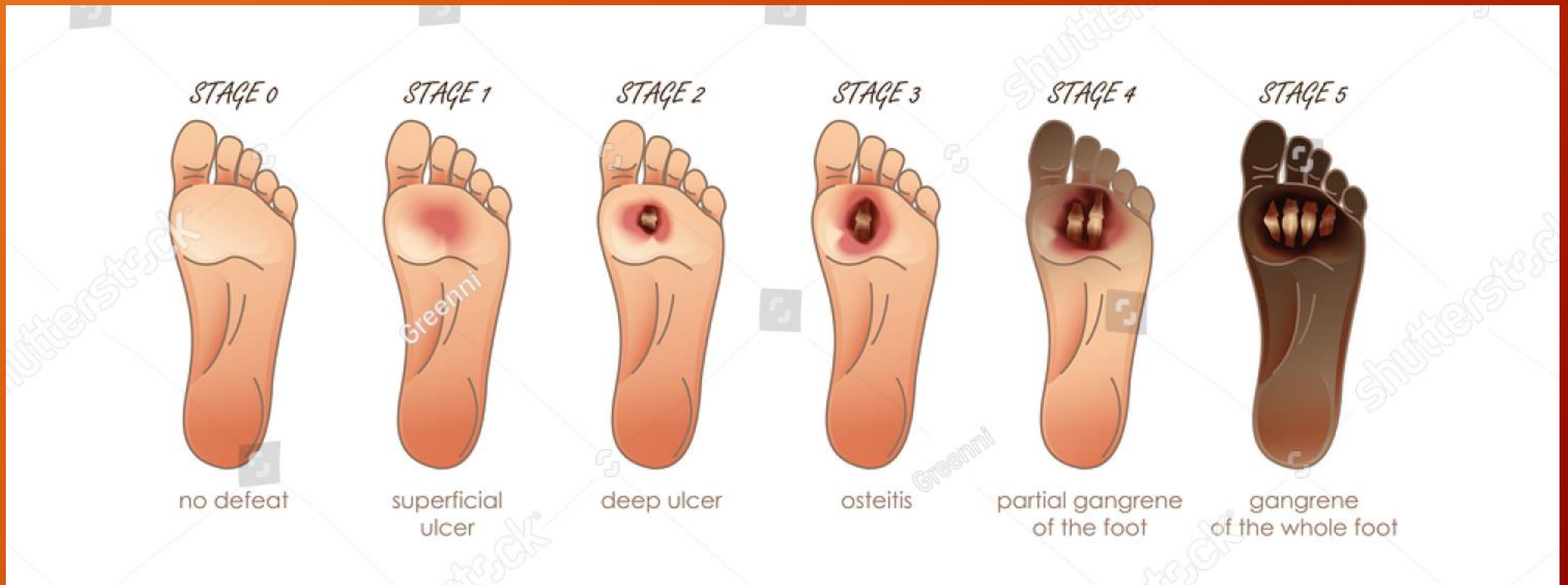
- ◆ Lifetime risk of up to 34% in patients with diabetes
- ◆ Account for approximately 2/3 of all non traumatic amputations performed in the US.
- ◆ Infected or ischemic diabetic foot ulcers account for approx 25% of all hospital stays for patients with DM

What do you do?

Your goal when assessing a diabetic ulcer? Determine the following:

1. Extent and depth of ulcer:
2. Presence of infection and/or ischemia: ORDER AN ANKLE-BRACHIAL INDEX + toe pressure measurements*

(Old) Staging of Diabetic Ulcers: the Wagner System



(New) Staging of Diabetic Ulcers: the University of Texas System → Grades and Stages

Grades
0: Pre- or post-ulcerative
1: Full-thickness (FT) ulcer that does not involve the underlying tissues
2: FT ulcer with tendon or capsular involvement, NO PALPABLE BONE
3: FT ulcer through to the bone



Stages
A: non-infected
B: infected
C: ischemic
D: Infected and ischemic

Grade 0: Pre- or Post-ulcerative

Superficial diabetic foot ulcer



Foot from a diabetic patient showing a superficial ulcer that involves a partial thickness of the skin but no involvement of the underlying tissues. This lesion healed quickly with rest and local foot care.

Courtesy of David McCulloch, MD.

UpToDate®

Grade 1: Full-thickness

Full thickness diabetic foot ulcer



Foot from a diabetic patient with a penetrating neuropathic ulcer that is not associated with abscess formation or bone involvement. The toes have begun to claw (ie, extension deformity at the metatarsal phalangeal joint and flexion deformity at the proximal interphalangeal joint) because the pull of the extensor digitorum longus, an extrinsic muscle in the anterior compartment of the leg, overpowers the neuropathy-induced weakness of the intrinsic foot muscles. The proximal phalanges will eventually sublux, or even dislocate, dorsally. As the deformities progress, the metatarsal fat pad will migrate distally and the toes will lose the ability to effectively plantarflex and offload the metatarsal heads. The combined effect increases pressure under the metatarsal heads, predisposing to ulcer formation.

Courtesy of David McCulloch, MD.

Grade 2: Tendon or Capsular Involvement



Grade 3: Visible or Probes to bone



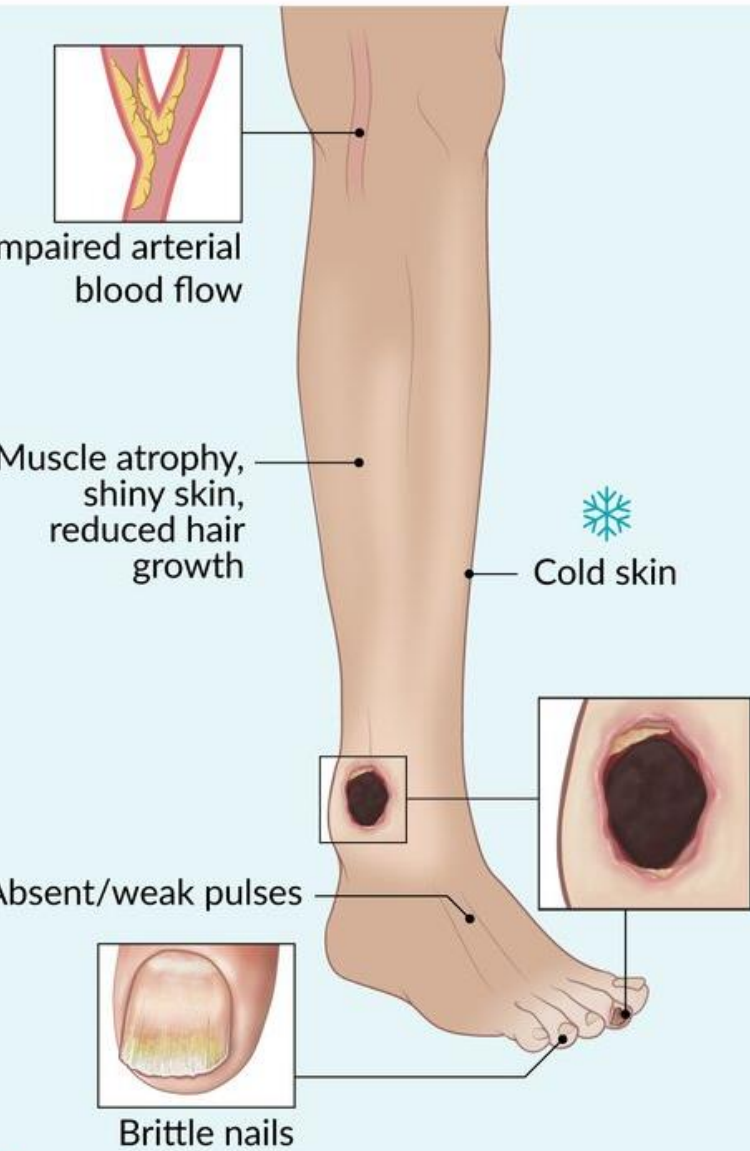
Diabetic Ulcer Treatment

- ◆ Reduce tissue pressure - mechanical offloading
 - ◆ i.e. Bed rest, total contact casts (fiberglass distributes weight throughout entire foot), wedge shoes (relieves pressure from heel or back of foot)
 - ◆ Helpful for your patient to ambulate as much as possible to maintain strength and adequate perfusion
- ◆ Maintain a moist wound environment → Moist >> Wet >> Dry: Moist and protected wounds have the best shot at healing well and effectively.
- ◆ Debride wound
- ◆ Treat infections
- ◆ Control pain, adequate nutrition
- ◆ Surgery if needed
- ◆ *Keep track of wound progress! Measure it at each visit to track healing rate.*
 - ◆ If you've had an ulcer, you are at higher risk for recurrence! Think of it as "being in remission".

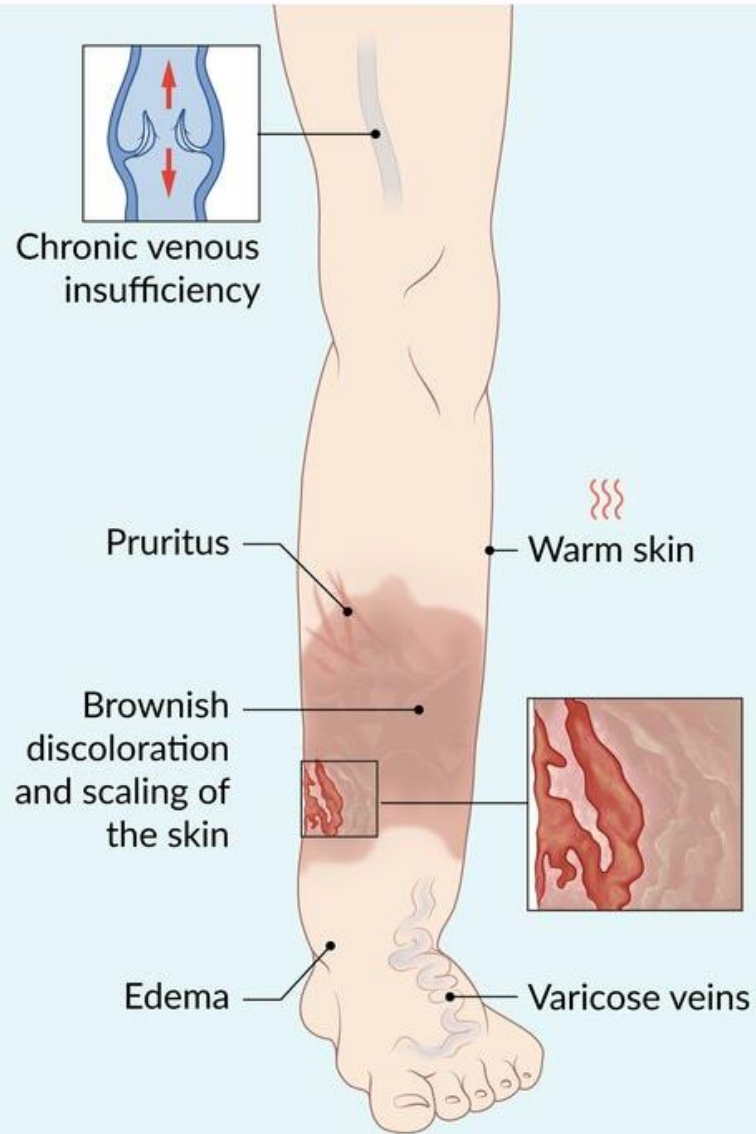


Figure 2. Progression of autolytic debridement with self-adaptive wound dressings (SAWDs): (A) Case 2 before treatment with SAWD; (B) day 2 of treatment with SAWDs and compression wraps; and (C) day 21 of treatment.

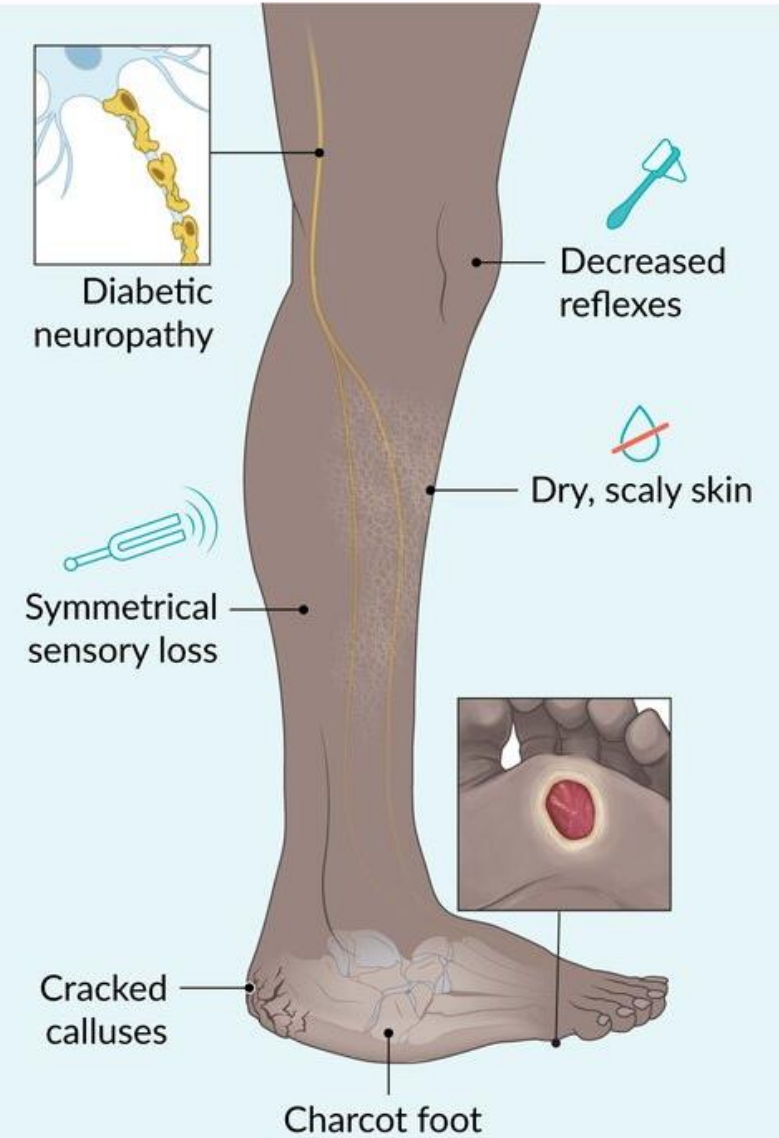
Arterial ulcer



Venous ulcer



Neuropathic ulcer



Very painful

Painless





Unna Boot

- ◆ Gauze impregnated with zinc paste, calamine lotion and glycerin; hardens
- ◆ Commonly applied for the treatment of venous stasis ulcers
- ◆ Promotes healing, increases blood return to the heart and reduces infection



Hyperbaric Oxygen Therapy

- ◆ Breathing pure oxygen in a pressurized room or tube
- ◆ Oxygenation of your blood increases which can stimulate growth factors and promote healing
- ◆ Risks of temporary myopia, middle ear injuries, lung collapse, seizure
- ◆ May require 20-40 treatments



Excessive Healing

- ◆ Hypertrophic scars and keloids: overabundance of fibroplasia in the healing process
 - ◆ Hypertrophic scarring is more common in lightly-pigmented skin
- ◆ Keloid formation is 15x more common in darkly-pigmented skin

END